

Imaging the Musculoskeletal System Lecture



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Objectives

- Principles and Techniques used in Musculoskeletal Imaging
- Vocabulary for Describing Bony Anatomy
- Fracture and dislocation Language, how to communicate information about a fracture
- Language to Describe Bone Lesions
- Recognize various osseous pathology on radiographs
- Periosteal changes, appearance and their significance



Basic approach to bone films

- Systematic approach
 - Most important- Identifying the body part 😊
 - Identifying the views
 - The standard views will be at **minimum** frontal and lateral views
 - Most joints will also include an oblique view/oblique variant
 - Always evaluate if the patient is an adult or a child as your radiographic differential diagnosis will change based on the age of the patient

Basic approach to bone films

- Step 1
 - Checking alignment of the osseous structures
- Step 2
 - Evaluate joint spacing
 - Do they appear narrowed or widened? How does the joint in question compare with other joints visible on the film?
- Step 3
 - Evaluate the bone cortex it should be smooth and continuous
 - Nutrient foramina- long bone via an obliquely oriented canal. They are often observed with a well-corticated edge
 - Physeal plates and sesamoids
- Step 4
 - Evaluate the surrounding soft tissue structures!!
 - Is there soft tissue swelling? Joint effusion?

Maximum x-ray
Transmission
(least dense
tissue)

Darkest

air

fat

soft tissue/fluid*

calcium

bone

x-ray contrast

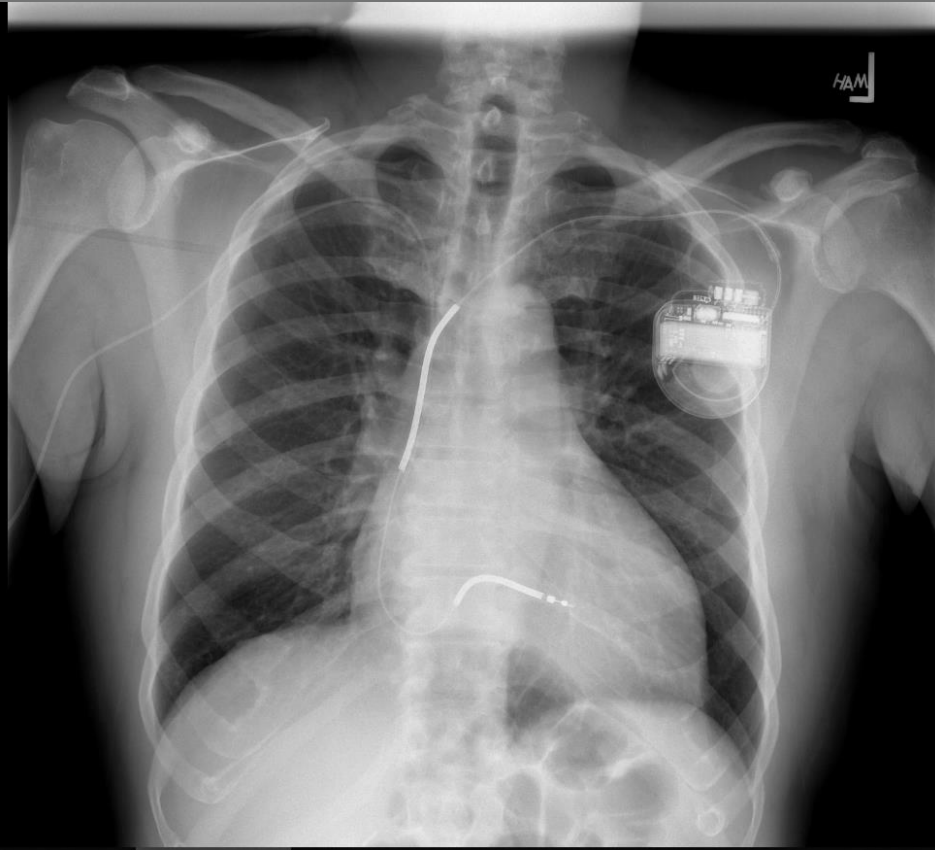
metal

Maximum x-
ray

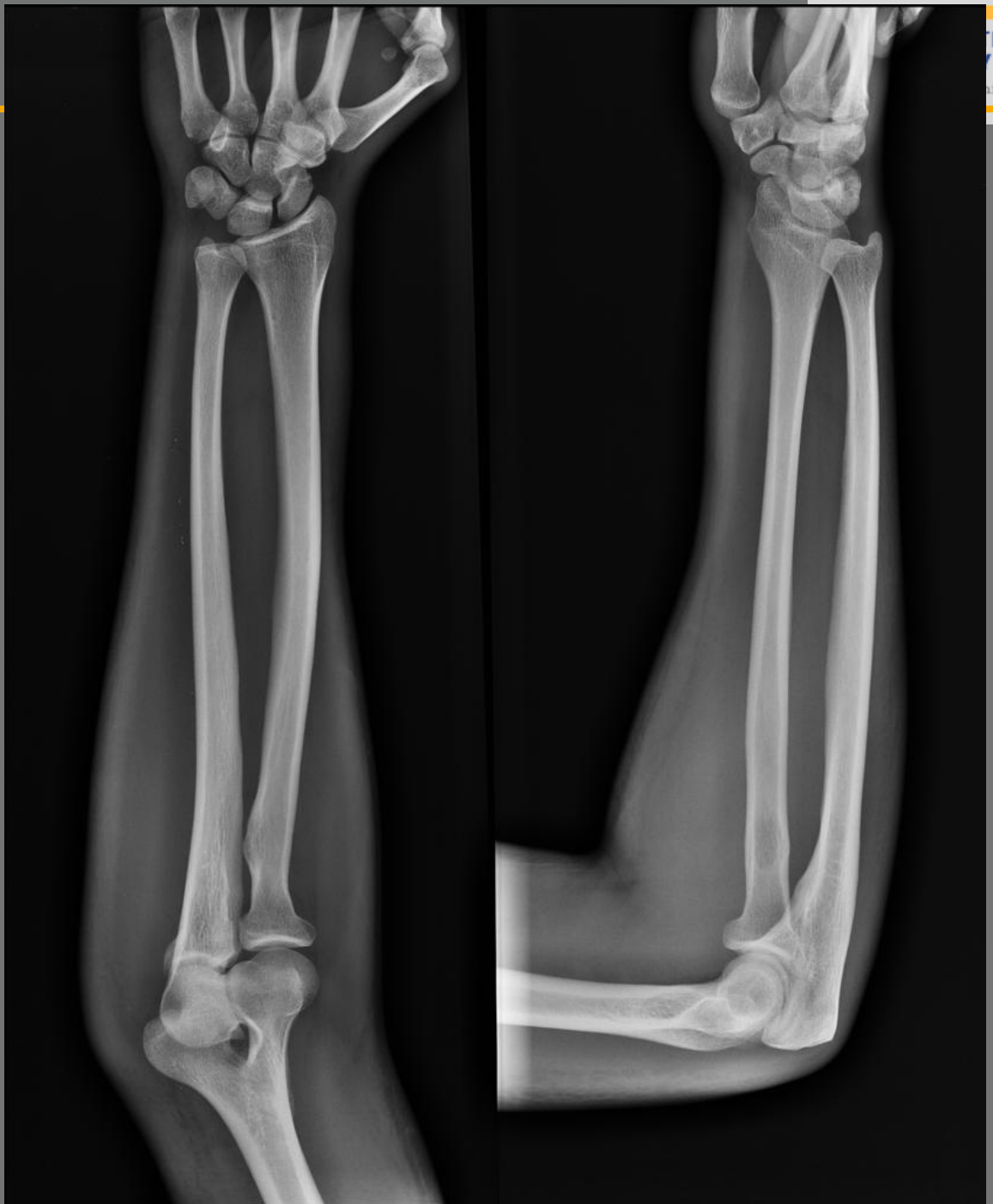
Whitest

Absorption
(densest tissue)

*fluid can be pus, blood or
simple fluid



- For example:
Can you identify the muscles of the forearm on this radiograph?



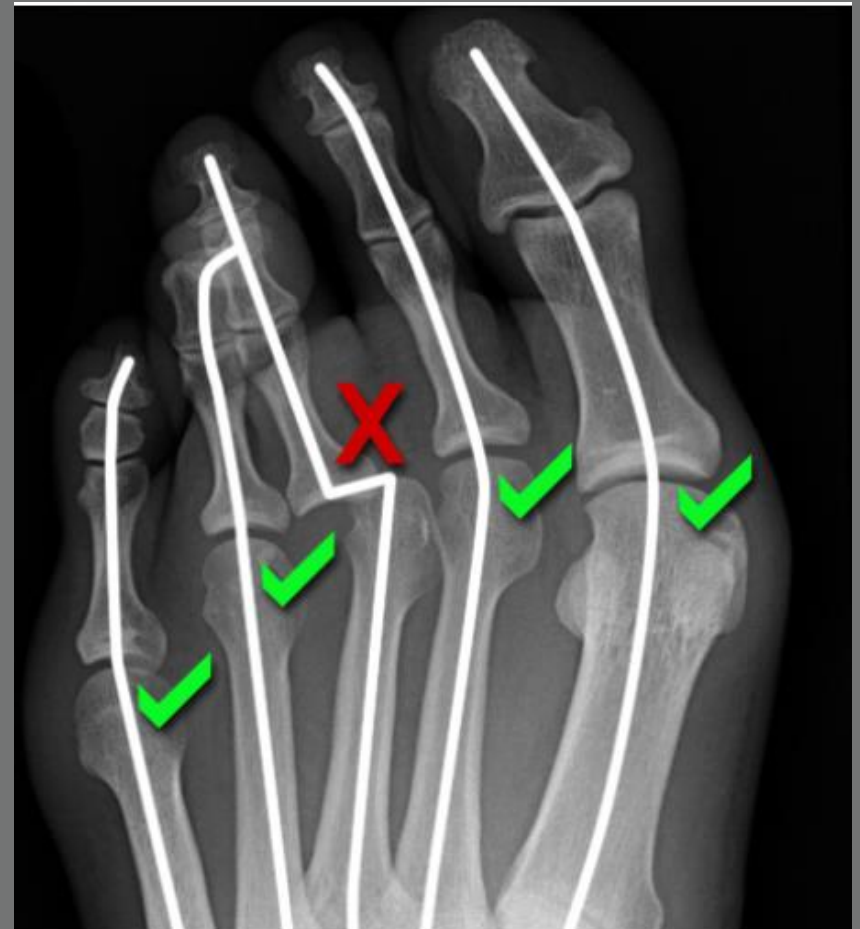
Joint alignment

- Loss of alignment may be due to a fracture or a joint dislocation.



Joint alignment

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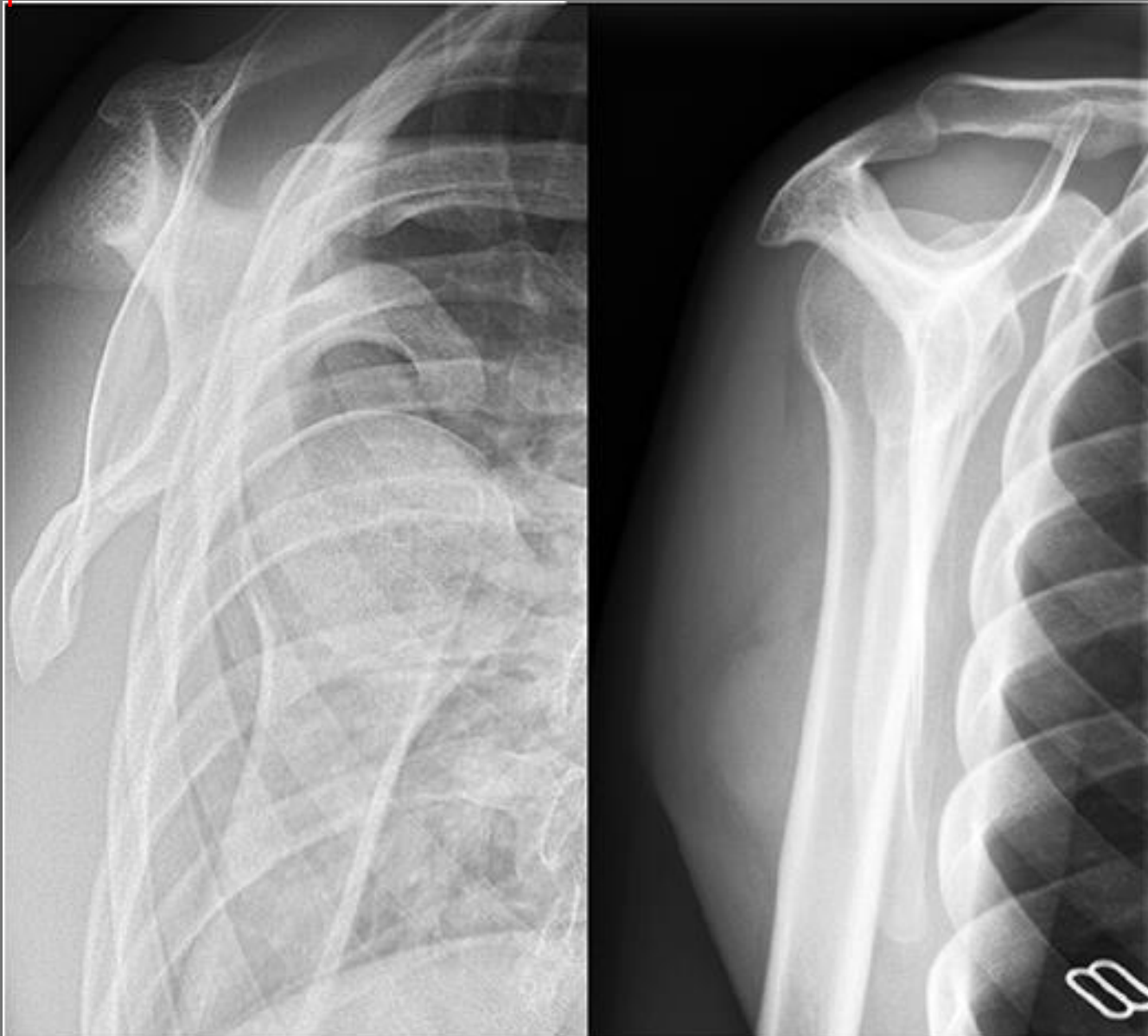
Anterior shoulder dislocation
humeral head beneath the
coracoid process



Normal

Anterior shoulder
dislocation humeral head
beneath the coracoid
process**

Normal



Joint spacing

- Maybe narrowed due to cartilage loss or widened due to dislocation/dissociation.



Cortical outline

- Check the cortical outline of ALL bones visible on every X-ray available
- Check for any breach in this outline
- Do not stop when you find one abnormality
- keep going until your eyes have covered all bones



Cortical outline

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- Check for any breach in this outline
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Nutrient Foramina*



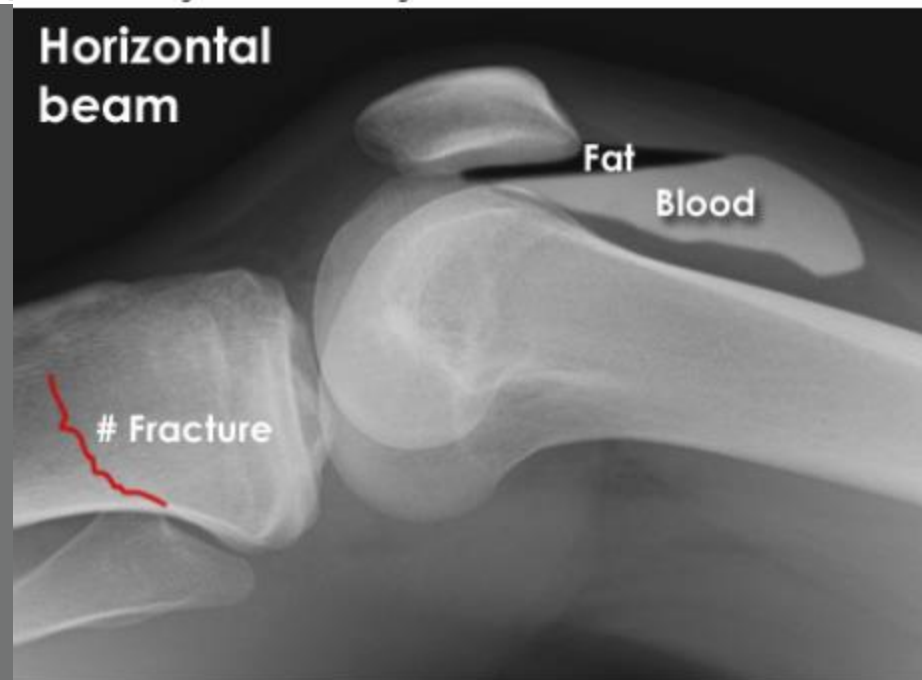
Soft tissues *

- Not uncommonly an abnormality of soft tissues is more obvious than a bone injury, or may even imply a bone injury that is not visible at all.
- Look for focal soft tissue swelling as well as the “joint spaces” themselves for effusions



Soft tissues

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Viewing principles

- At least 2 views of the body part in question are usually required
- Satisfaction of search!

Viewing principles

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Viewing principles



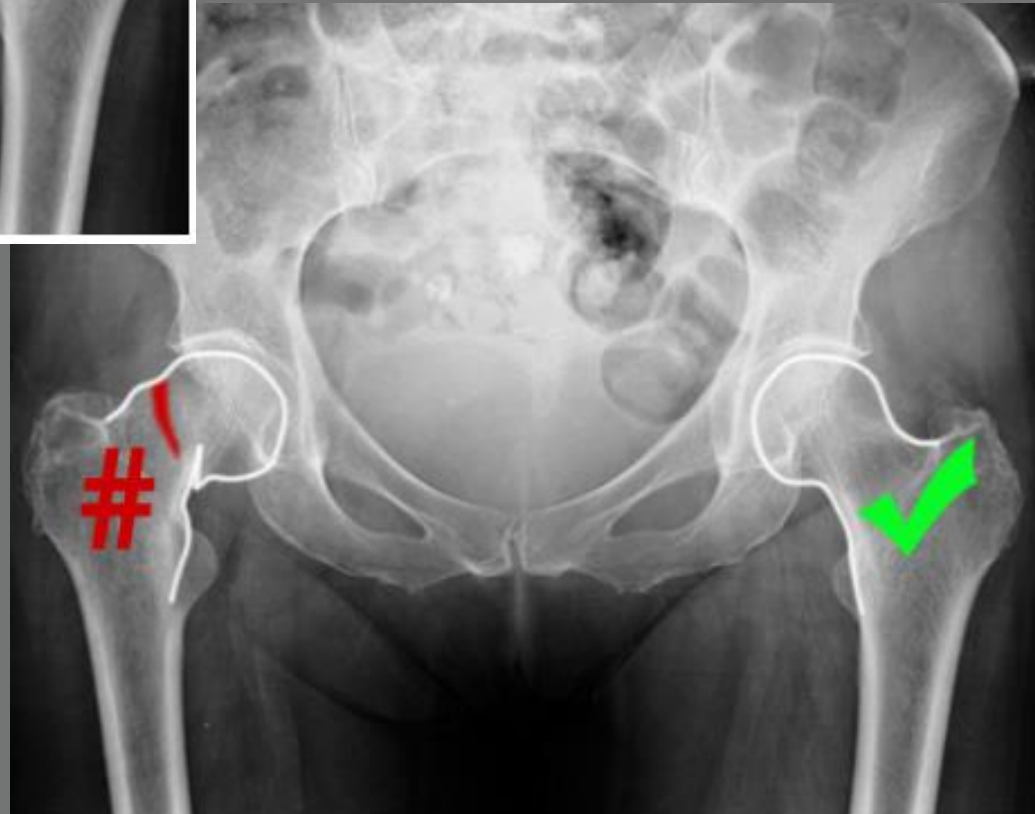




Compare sides



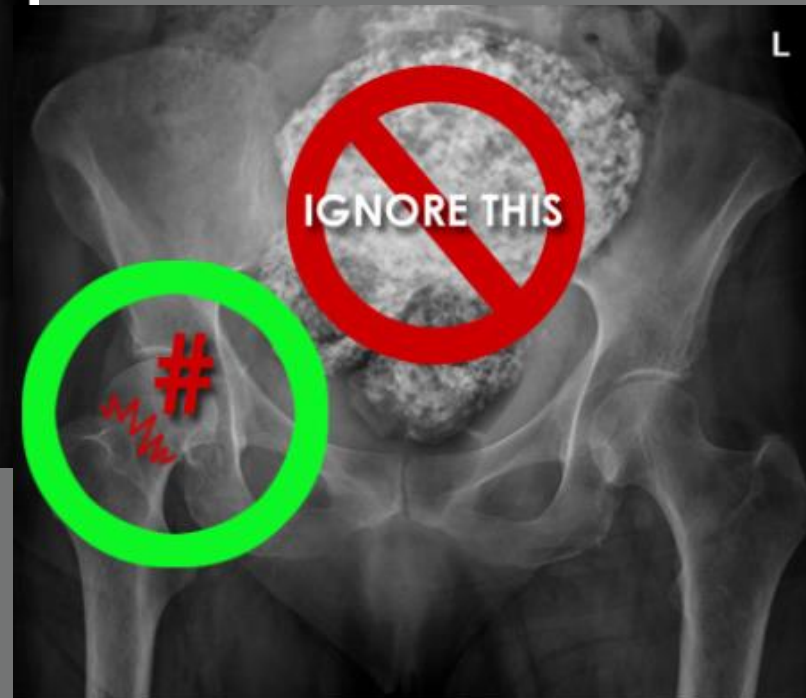
Compare sides



Satisfaction of search



Satisfaction of search



Evaluate the whole film



Evaluate the whole film





Im: 1,661



Im: 1,662





Fracture Description

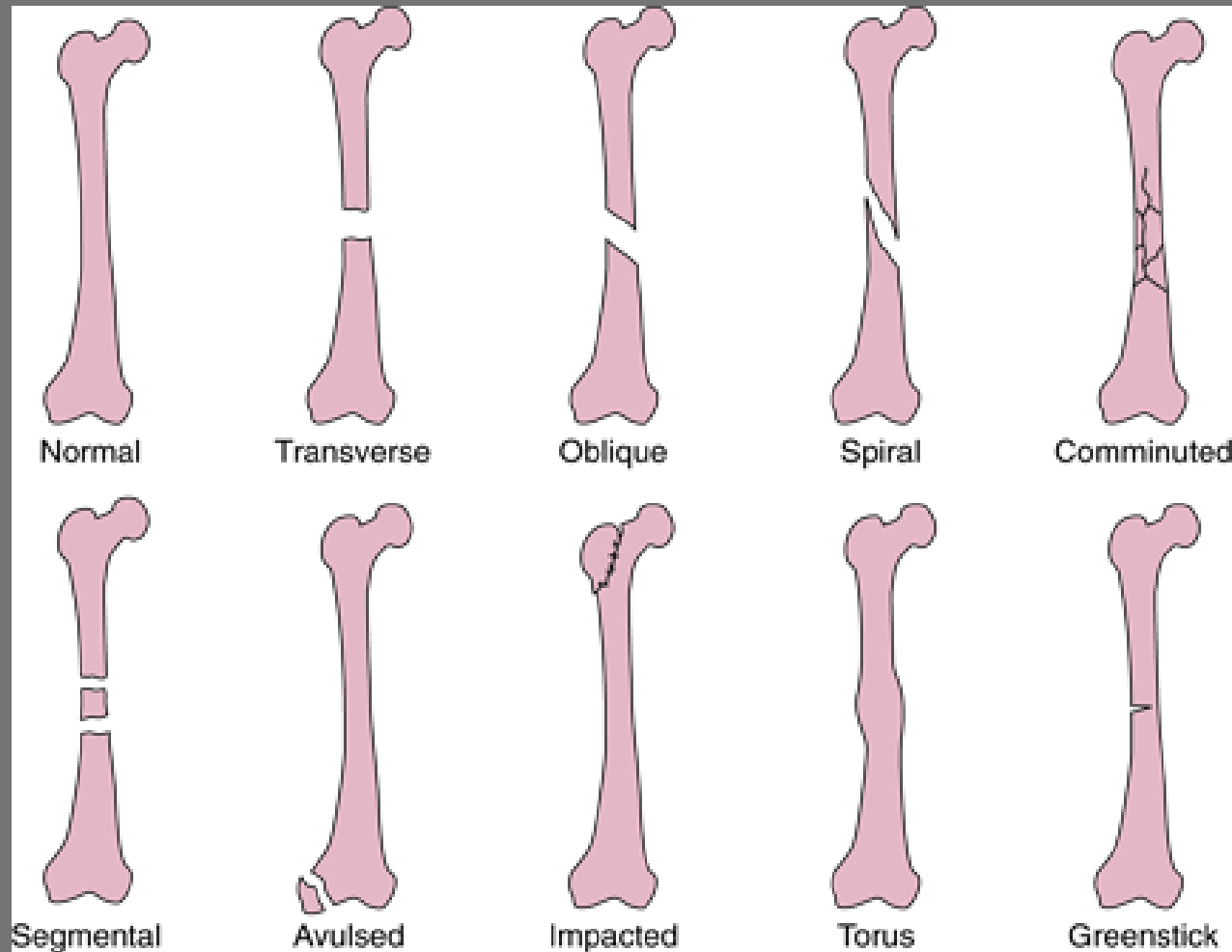
- **Direction** of fracture line:
 - Transverse
 - Oblique
 - Spiral
 - Longitudinal
- **Alignment of fracture: Displacement-** named for distal fragment
 - Angulation
- **Comminution**
- **Articular Involvement**

Fracture Description

- **Open vs Closed:** fracture is open when exposed to air (laceration or gross exposure)
- **Pathologic fracture:**
 - Normal stress on abnormal bone
 - Implies fracture through weakened bone i.e a bone lesion, severe osteopenia
- **Stress fracture:**
 - Implies misuse or overuse. Abnormal stress on normal bone

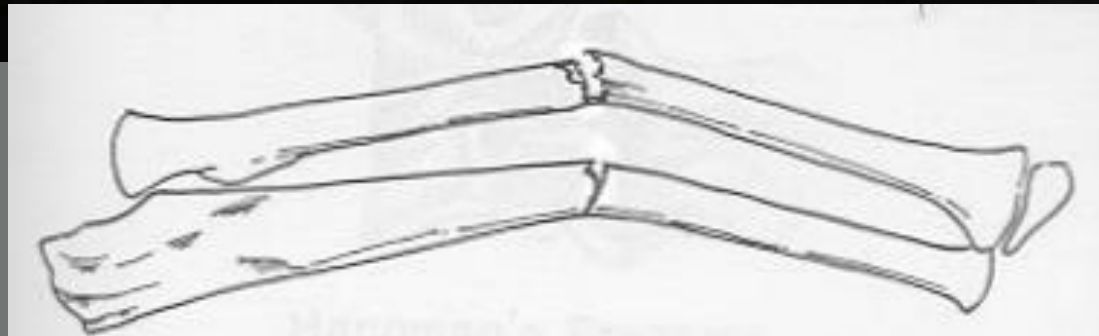


Path of the Fracture



Greenstick fracture

- Cortex broken on one side of the bone and only bent or buckled on the other side



Stress Fracture



Stress Fracture



Patterns of Bone Growth and Destruction- Bone Lesions



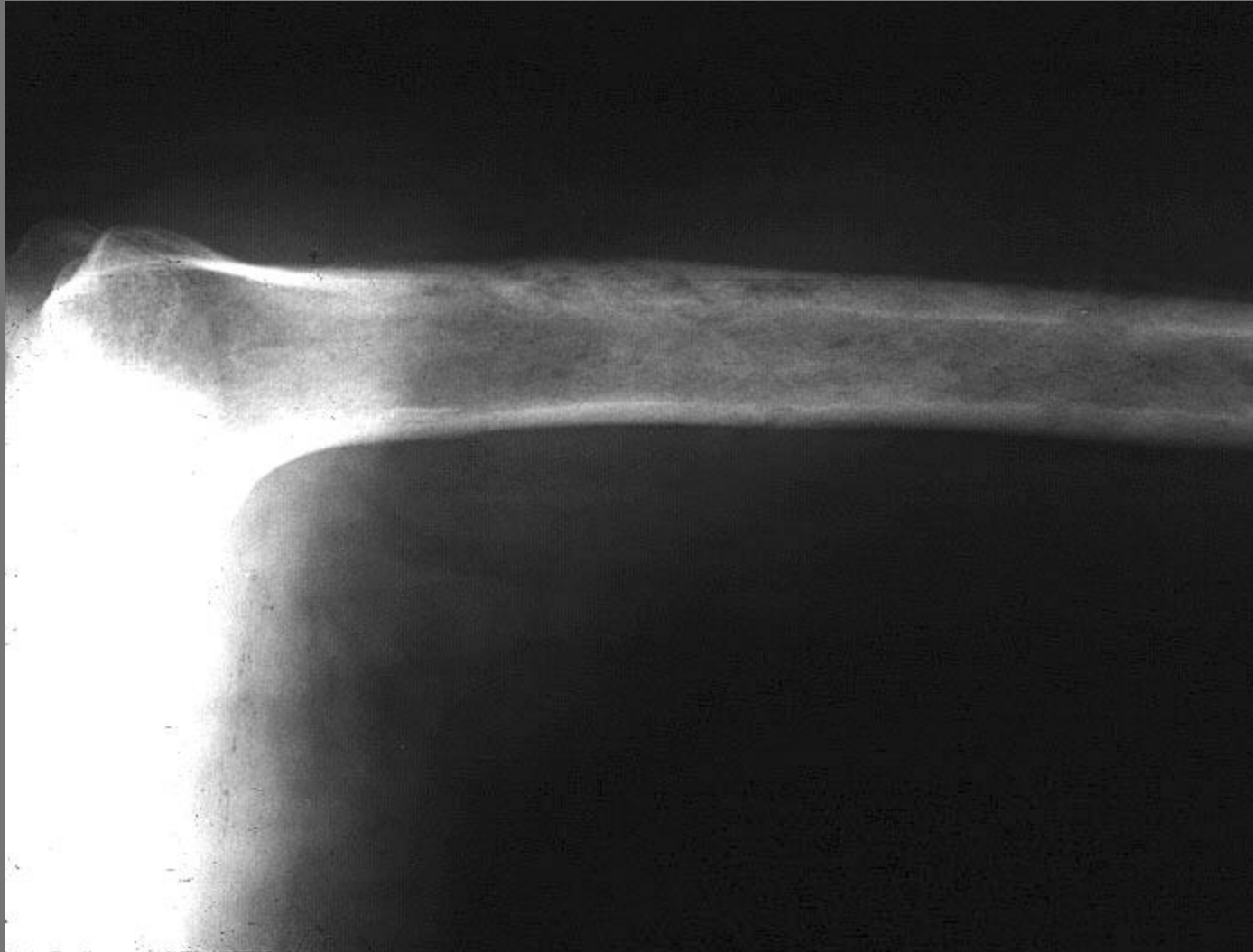
- Benign and non-growing (or extremely slowly growing) lesions are well circumscribed and show **geographic pattern** of bone destruction with a **sclerotic rim**.
 - Geographic pattern refers to a **well-defined** area of lysis.
 - Well defined in that I can take a pencil and trace around the lesion.
 - The sclerotic rim is more commonly seen in the weight-bearing bones and represents bone reaction to the lesion.
 - Its presence means that the bone has been given sufficient time to react.
 - Some authors say that the sclerotic rim signifies benignancy to about 95%.

- If the lesion is growing more rapidly, it may still show a well-demarcated zone of bone destruction (geographic pattern), but it will lack a sclerotic rim.
- With continued growth, such lesions may show **cortical expansion**.
- Expansile growth pattern is defined as visible widening of the affected portion of bone.
- This pattern may be seen in locally aggressive tumors and in malignancies.

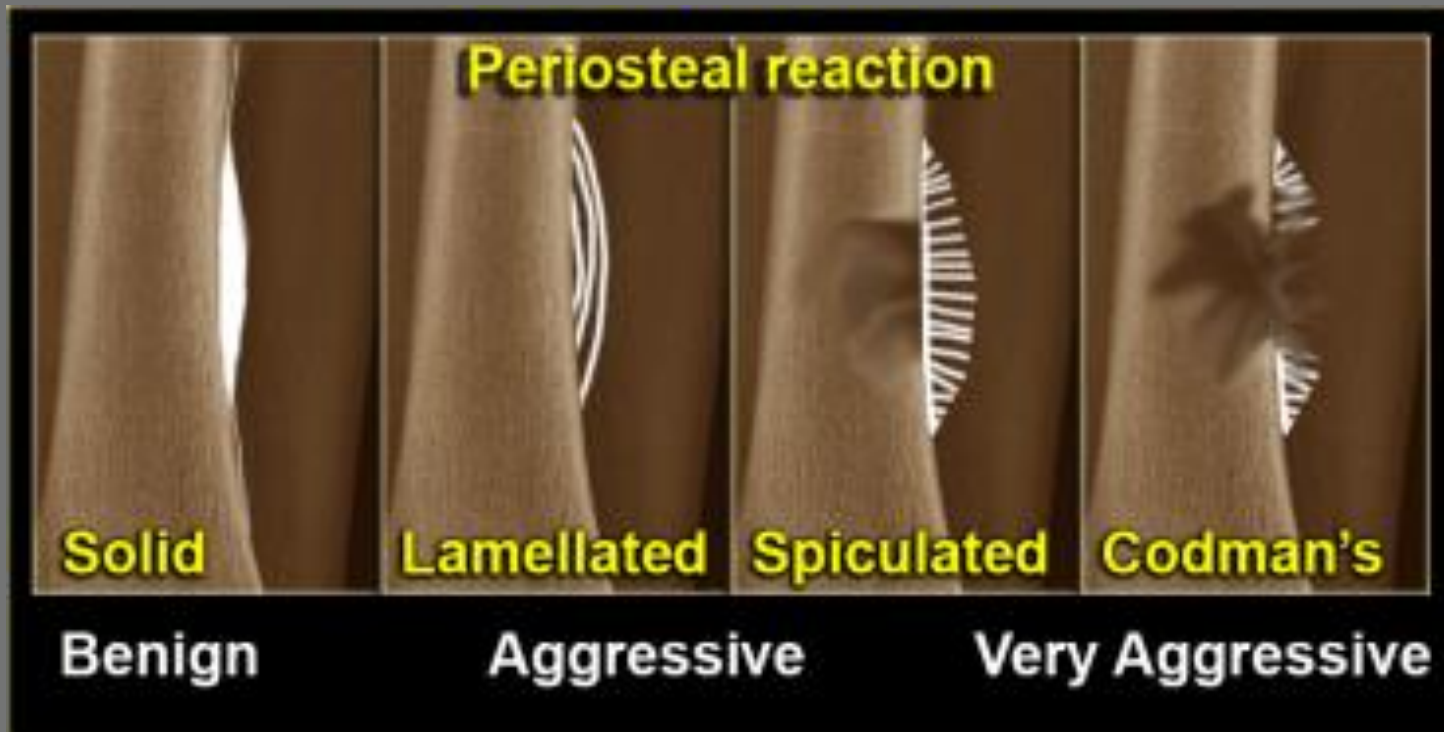




- Rapidly growing lesions are poorly defined and may show aggressive, infiltrative patterns of bone destruction (**permeative or "moth-eaten"**).
- **"Moth-eaten" pattern** is defined as an ill-defined zone of multiple small radiolucencies that may coalesce.
- They are seen in **high-grade malignant neoplasms and in osteomyelitis**



Periosteal Reactions



Periosteal reaction may be classified as benign or aggressive (note: not benign and malignant) based on the time course of the initiating process.

Periosteal Reactions



Slow-growing tumors
provoke **focal cortical
thickening** (solid
periosteal reaction, or
"buttress")

Periosteal Reactions



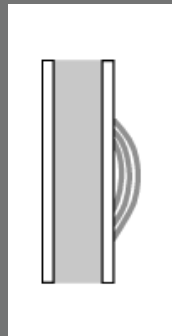
- Slow-growing processes, the periosteum has plenty of time to respond to the process.
- That is, it can produce new bone just as fast as the lesion is growing.
- Therefore, one would expect to see solid, uninterrupted periosteal new bone along the margin of the affected bone solid periosteal reaction along the cortex of a bone



Periosteal Reactions

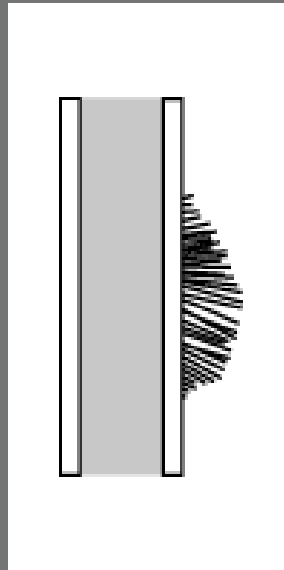


- Rapidly growing processes, the periosteum **cannot** produce new bone as fast as the lesion is growing.
- Therefore, rather than a solid pattern of new bone formation, we see an interrupted pattern.
- This interrupted pattern can manifest itself in several ways, depending on just how steadily the lesion grows.
- If the lesion grows sporadically, then the periosteum may have time to lay down a thin shell of calcified new bone before the lesion takes off again on its next growth spurt.
- This may result in a pattern of one or more concentric shells of new bone over the lesion.
- This pattern is sometimes called lamellated or “onion-skin” periosteal reaction.



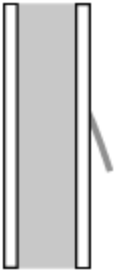
Periosteal Reactions

- If the lesion grows rapidly but steadily, the periosteum will not have enough time to lay down even a thin shell of bone, and the pattern may appear quite different.
- In such cases, the tiny fibers that connect the periosteum to the bone (Sharpey's fibers) become stretched out perpendicular to the bone.
- When these fibers ossify, they produce a pattern sometimes called “sunburst” or “hair-on-end” periosteal reaction, depending of how much of the bone is involved by the process.





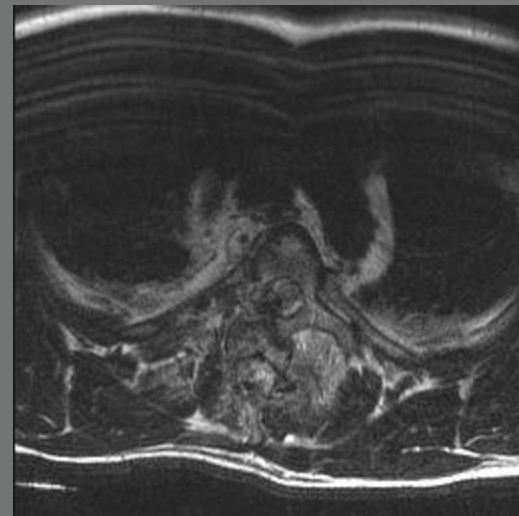
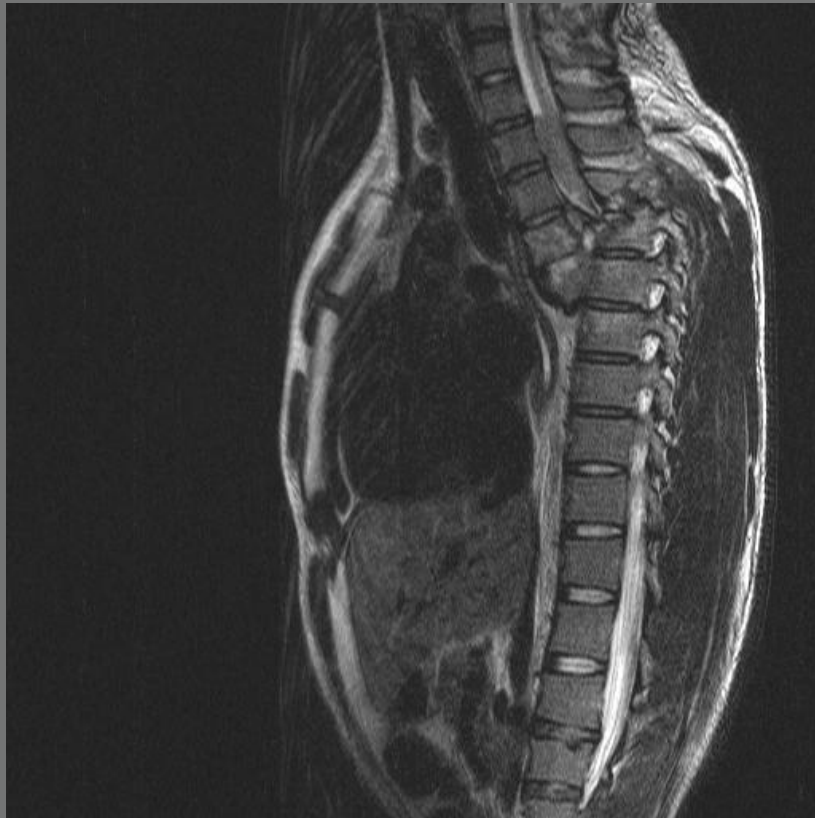
Periosteal Reactions



- Another pattern seen in rapidly growing processes is called the **Codman's triangle**.
- This is a bit of a misnomer, since there really is not a complete triangle.
- The process is growing too fast for the periosteum to respond
- When this little bit of ossification is seen tangentially on a radiograph, it forms a small angle with the surface of the bone, but not a complete triangle.
- So, when a process is growing too fast for even the Sharpey's fibers to ossify, one may only see a soft tissue mass arising from the bone, perhaps with small Codman's triangles at its margins.
- <https://rad.washington.edu/about-us/academic-sections/musculoskeletal-radiology/teaching-materials/online-musculoskeletal-radiology-book/periosteal-reaction/>



Codman's
Triangle



<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>